Helena Fu

(650) 863 - 4937 | Helena.Fu@tufts.edu | linkedin.com/in/helena-fu-ml | github.com/helena-f

Education

Tufts University - School of Engineering — Medford, MA

B.S. in Computer Science, B.S. in Mathematics (GPA: 3.93 / 4.00) — *May 2027*

Honors: 4x Tufts Dean's List, Al Gardner Memorial Endowment Scholarship (Society of Women Engineers)

Relevant Coursework: Machine Learning Foundations, Linear Algebra, Differential Equations, Data Structures, Statistics, Real Analysis, Mathematical Aspects of Data Analysis, Algorithms, Mathematical Modeling, Statistical Pattern Recognition

Technical Skills

Machine Learning/AI: Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, OpenAI, Claude, Gemini, Cursor
Languages: Java, Python, C/C++/C#, SQL, JavaScript, Typescript, HTML/CSS, R, Assembly, ROS, Bash Script, MatLab, PHP
Frameworks: React, Node.js, Flask, WordPress, Next.js
Developer Tools: Git, Github, Docker, Google Cloud Platform, Visual Studio Code, IntelliJ, AWS Bedrock, AWS Lambda, Figma
Libraries: pandas, NumPy, Matplotlib, HuggingFace

Experience

Mad Realities — New York City, NY

Software Engineering Intern — *June 2025 – Present*

- Built LLM interface using **Next.js**, **Gemini, Pinecone DB,** and **LangChain** to streamline creative process for producers.
- Integrated Retrieval-Augmented Generation **(RAG)** system trained on **10K+** media segments to enable script retrieval.
- Improved semantic search accuracy by **85%** through embedding optimization, prompt engineering, and RAG fine-tuning.

Acadia Analytics LLC — Jan. 2025 – May 2025

Founding Quantitative Analyst — *Medford, MA*

- Engineered automated trading signals with **BERT, RL, LSTM,** and **HMM**; increased ROI by **53%** over baseline strategies.
- Developed real-time inference and retraining pipeline in Python to adapt models to live market data.
- Formulated and tested alpha hypotheses by **analyzing sentiment** from financial news and price movement correlations.

Human-Robot Interaction Lab at Tufts University — Medford, MA

Research Assistant — *Feb. 2024 – May 2025*

- Integrated **NLP system** with DIARC robot architecture to enable unstructured commands from **50+** study participants.
- Implemented **natural language cache** to store and retrieve prior LLM responses, reducing latency by **46%**.
- Built real-time pupillometry preprocessing pipeline in Python; boosted data quality for cognitive state ML model by **30%**.

MIT Schwarzman College of Computing — Cambridge, MA

Machine Learning Intern — *Aug. 2024 – Dec. 2024*

- Developed **multi-modal ML pipeline** for hydrographic prediction; achieved **97%** regression accuracy.
- Engineered features and applied ensemble models to predict harmful algal bloom risk with **76%** accuracy.
- Preprocessed **11GB** of unstructured sensor and biochemical data from **3** marine data sources.

Projects

DSCVR.AI | *Cornell NYC AI **Hackathon Winner***

- Built AI-assisted writing platform with LLM-driven autocomplete; engineered seamless UX with **Next.js** and **Tailwind**.
- Posttrained foundation model using domain-specific prompts via **Claude API**, **Docker**, and **AWS Bedrock**.
- Earned **Best UX**, and **Most App Store Ready** awards for integrated ML experience.

AJL Kaggle Competition | ***6th** Place out of **350+***

- Evaluated Vision Transformer variants (DeiT, Swin, Google ViT, ConvNeXt V2) via **TorchVision** on Fitzpatrick17k dataset.
- Achieved **71%** on image classification task with custom **PyTorch** training pipeline using **CrossEntropyLoss** and **Adam**.
- Collaborated with team of 4; awarded top prize for novel image preprocessing algorithms improving model robustness.

Tufts JumboCode | *Developer for A2Empowerment Team*

- Built website with React, Node.js, MongoDB, **AGILE** management, and **REST API**s to organize **150+** volunteers.